

Considerations for Building Credible, Fair and Trustworthy Machine Learning Models for Biomedical Engineering

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May 4, 2026



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Research Council of Canada

Conseil de recherches en sciences
naturelles et en génie du Canada

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In Short

- Machine learning (ML) algorithms can be very effective in biomedical signal processing. However, their utility depends on following good methods of training and testing.
- We present practical guidance to help researchers achieve consistent model reliability.
- Enhancing reliability of predictions in clinical applications will improve patient care and safety.
- The proposed recommendations are practical and will enable developers to increase credibility and reliability of ML models being used.

Background

- ML can process biomedical signals in higher dimensionality and greater capacity [1], but model development requires specialized expertise.
- Practical guidelines on how to build them are lacking [1] and gradually emerging [2,3].
- We present recommendations on how to build ML models tailored for biomedical signal processing applications to improve credibility, trustworthiness of prediction models.

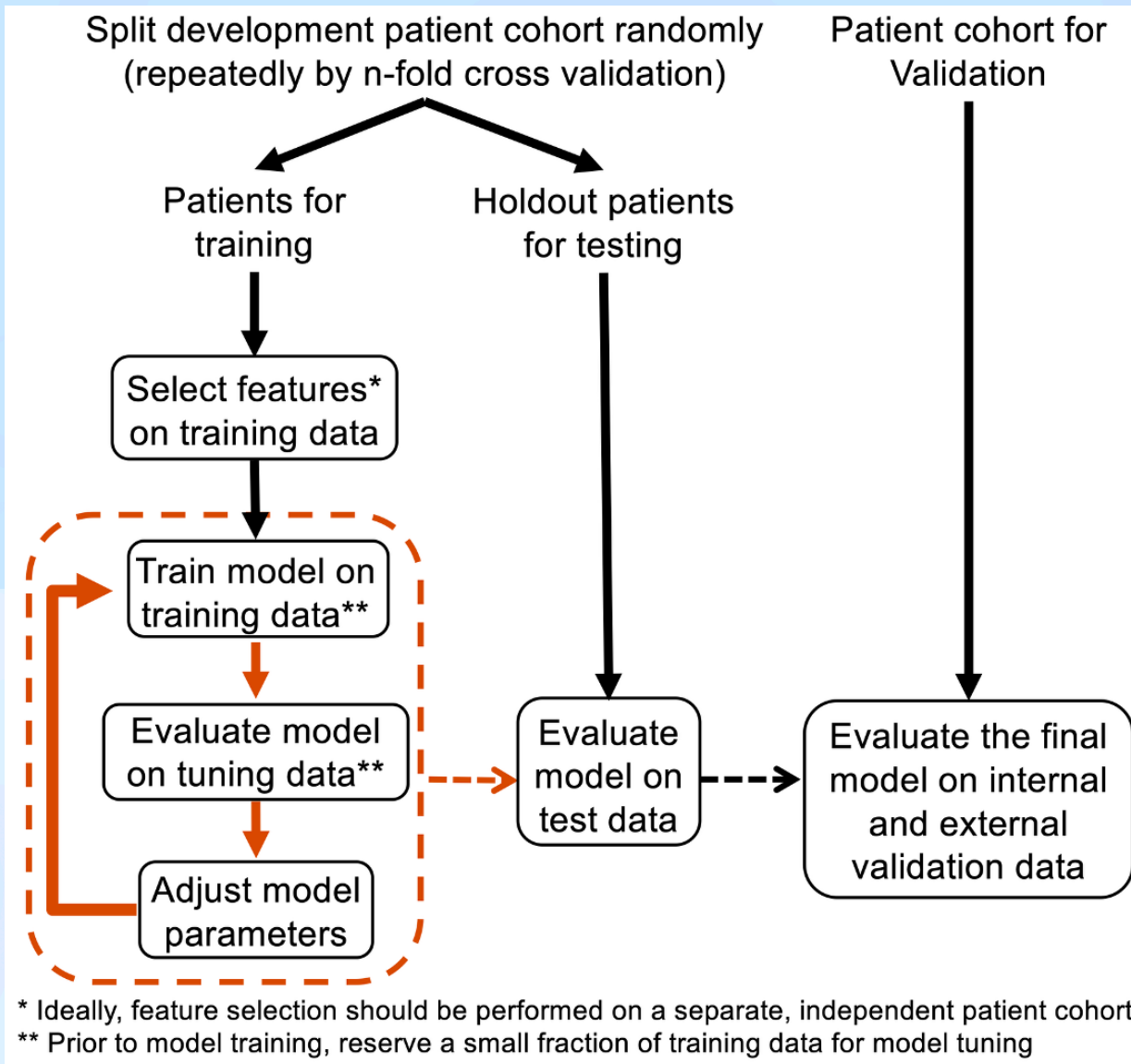
Methodology

- $D=\{x_i, y_i\}$ is a dataset of n data points of m features
- $X = x_i$ are features and $Y=y_i$ are outcomes for $i=1,2,\dots,n$.
- A model $f: X \rightarrow Y$ may be a classification or regression.
- ML algorithms assume (x_i, y_i) are independently drawn from the same joint probability distribution $P(X, Y)$ where:

$$P(X, Y) = P(Y | X) * P(X)$$

- The above is rewritten as i.i.d assumptions [5]:
 1. (x_i, y_i) is **independent** of (x_j, y_j) for all $i, j=1,2,\dots,n$ and $i \neq j$
 2. (x_i, y_i) is **identically distributed** as $P(X, Y)$ for all $i, j=1,2,\dots,n$

- In biomedical applications, particularly in signal processing, the above are often violated
- Violating 1 can introduce unintended consequences (bias) and / or data leakage [4]
- Violating 2 will increase prediction errors [5].
- When building an ML pipeline, we recommend the experiment the following design as supported by [3,6].



Discussion

- **Data leakage** occurs when information external to training data is used for building the model (violates assumption 1).
- ML models should only be **tested on unseen data** from patients, i.e., not included in training.
- Feature selection, model tuning, adjustment and fitting should only be carried out only on training data [4].
- The final model should be **validated on unseen subjects** drawn from the same institution (internal) or from other centers (external) to ensure compliance with assumption (Institutional variations may limit model generalizability).

Conclusion

- **Reliability of prediction performance is crucial for trustworthy evaluation to achieve credibility**
- **Unnecessary and potentially overwhelming false positives increase the burden for healthcare applications**

References

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